

II. World Geography

The Universe

i. General Concept

*The first rational perspective regarding the origin of the planet was probably proposed by a French scientist 'Comte de Buffon' in 1749. According to the number of stars participating in the origin of planets, the scientific concepts have been divided into two parts- (1) **Monistic concept** and (2) **Dualistic concept**.

*Under the Monistic concept, Immanuel Kant's Gaseous Hypothesis and Pierre Simon Laplace's Nebular Hypothesis became famous. Under the Dualistic concept, Chamberlin and **Moulton** Planetesimal Hypothesis, James Jeans and Jeffery's Tidal Hypothesis, H.N. Russell and Littleton Binary Star Hypothesis etc. became noteworthy.

*The latest theory regarding the origin of the universe is the **Big Bang Theory**. ***George Lemaitre** (1894-1966) was the first to present the idea of an expanding universe. Later in 1967, Robert Wagoner presented the interpretation of this theory. The Big Bang theory explains how **13.7** or **13.8** billion years ago universe expanded from a very high density and high-temperature state. *After the explosion, the universe expanded in all directions and different galaxies of different shapes were formed. Thus, the universe comprises of an infinite number of galaxies.

It is estimated that there are appx. 100-200 billion galaxies in the universe and every galaxy is estimated to have **around 100 billion stars**. Our galaxy is known as an **Akash Ganga** or **Milky Way**.

*The shape of the Galaxies varies in shape and size. Some are spiral, some are irregular and some are elliptical. *In 2005, the Spitzer Space Telescope gave clear evidences that our galaxy (the Milky Way) is **barred spiral** in shape, hence disapproving the earlier theories. *The universe is a collection of infinite galaxies. Every galaxy has around 100 billion stars. These stars are formed as a result of the presence of hydrogen and helium gases in their unified form (Nebulae). The gravitational force keeps the gases and dust particles in the unified form. Velocity, high temperature and organizational processes are the factors behind the formation of a star. *The formation of the star passes through different phases. In the first phase, a **proto-star** is formed after that the star enters the second phase i.e., the **main sequence Star**.

A star of the size of our sun stays in this phase for around 9 or 10 billion years. After this phase, the fuel of the star i.e., hydrogen starts getting used up to an extent that the star enters the third phase i.e., towards its **death**. *The lifespan of any star depends upon its size. A star of the size of our sun (one solar mass) has a lifespan of around 9 or 10 billion years. The bigger the star is, the shorter is its life span. For example, a star whose surface temperature is more than 30000 Kelvin and which is 100 times the size of our sun has a life span of only a few million years. After the main sequence life of the star ends, it enters in the elderly phase.

*In this phase, the outer layer of the star expands greatly, It becomes a **Red Giant** or **Red Super giant**. *After the red Giant or super Red Giant phase, a star witnesses an explosion in the form of a **Nova** or **Supernova**, thus entering its last stage of life. Depending upon its size it can end up in- a **Black Dwarf**, **Neutron star** or a **Black Hole**.

Sun like small stars – after the Red Giant phase and Nova explosion if the stellar mass is less than 1.44 solar Masses, the star will end up being a White Dwarf first and then into a Black Dwarf.

Medium sized stars - after the Red Giant Phase and Supernova explosion, if the stellar mass is between 1.44 to 3 solar masses, then a star can convert into a Neutron Star (Pulsar).

Large size stars – those which have a stellar mass more than 3 times that of our sun. These stars have the possibility to convert into a black hole after a supernova explosion. *In the year 1931 A.D. Indo-American Physicist **Subrahmanyan Chandrasekhar** invented the Chandrasekhar Limit. The Chandrasekhar limit which is appx. **1.44** times the mass of the sun (1.4 solar mass) determines the death state of a star. Any white dwarf with less than 1.4 solar mass will remain a white dwarf forever, while a star that exceeds this mass is destined to end in life in the most violent of explosions: a supernova, creating neutron stars or black holes. *It is to be noted here that the Chandrasekhar limit is determined after the nova or supernova explosion of the star has taken place. Prima facie it is determined by the mass of the residue of the exploded star. *It is not certain that which of the main sequence stars leave a residual mass under or over 1.44 times of solar mass, after explosion. However it is estimated that stars, with 8 times the size of our sun leave a residue